

AKI — Intrinsic 2: Nephrotoxins



1. Aminoglycosides

WHAT YOU SEE

VANCO/AMINO shelf, '5-7 days', cortex accumulation

WHAT IT ENCODES

Aminoglycosides (gentamicin, tobramycin, amikacin) cause non-oliguric ATN typically 5–7 days into therapy because they accumulate in proximal tubule cells via megalin-mediated uptake in the renal cortex. Risk rises with high trough levels, prolonged courses, volume depletion, and concurrent nephrotoxins. Once-daily dosing and trough monitoring reduce toxicity; classically also causes hypomagnesemia and hypokalemia.

BOARD TRAP

Choosing 'first dose' or 'immediate' AKI — aminoglycoside ATN has a characteristic 5–7 day lag from drug accumulation, and it is non-oliguric, so a stable urine output early on does not mean the kidneys are safe.

2. Vancomycin

WHAT YOU SEE

VANCO bottle on same shelf, synergy note

WHAT IT ENCODES

Vancomycin causes dose-dependent ATN, markedly worsened when combined with piperacillin-tazobactam or aminoglycosides (synergistic nephrotoxicity). Risk correlates with troughs/AUC; monitor levels and renal function. It can also rarely cause AIN. Hold or dose-reduce when creatinine rises.

BOARD TRAP

Blaming a rising creatinine solely on vancomycin in a patient also on pip-tazb — the combination (vanc + pip-tazo) is the high-yield synergistic nephrotoxic pair on boards, not vancomycin alone.

3. Contrast

WHAT YOU SEE

Radiation/contrast warning, '12-48 hours'

WHAT IT ENCODES

Contrast-induced nephropathy causes a creatinine rise at 12–48h post-contrast, peaking at 3–5 days and usually resolving within 7–10 days, via medullary vasoconstriction plus direct tubular toxicity. It is typically non-oliguric with a LOW FENa (<1%) despite being intrinsic ATN. Prevent with isotonic IV crystalloid (normal saline) before and after; highest risk in CKD, diabetes, volume depletion, and high contrast loads.

BOARD TRAP

Calling contrast nephropathy 'prerenal' because FENa is <1% — contrast ATN is one of the classic intrinsic causes that paradoxically gives a low FENa, so the timing (12–48h post-dye) and history, not FENa, make the call.

4. Amphotericin

WHAT YOU SEE

AMPHO B bottle

WHAT IT ENCODES

Amphotericin B causes dose-dependent direct tubular toxicity by inserting into tubular membranes and increasing permeability, producing AKI plus potassium and magnesium wasting and a distal (type 1) renal tubular acidosis. Liposomal formulations are less nephrotoxic. Pre-hydrate and replete K/Mg.

BOARD TRAP

Forgetting the electrolyte signature — students recall the AKI but miss that amphotericin classically causes hypokalemia, hypomagnesemia, and type 1 RTA, which is often the tested discriminator.

5. Cisplatin / chemo

WHAT YOU SEE

CHEMO shelf: cisplatin, ifosfamide, gemcitabine

WHAT IT ENCODES

Platinum agents (cisplatin) accumulate in the proximal tubule causing dose-dependent ATN plus magnesium wasting; prevent with vigorous saline hydration and amifostine. Ifosfamide causes proximal tubular damage with Fanconi syndrome (and hemorrhagic cystitis, prevented by mesna). Gemcitabine and mitomycin cause thrombotic microangiopathy (TMA).

BOARD TRAP

Lumping all chemo nephrotoxicity as 'ATN' — the board discriminators are ifosfamide → Fanconi, gemcitabine/mitomycin → TMA (schistocytes, low platelets), and cisplatin → ATN with hypomagnesemia.

6. Calcineurin inhibitor

WHAT YOU SEE

Tacrolimus/cyclosporine bottle, vasoconstriction

WHAT IT ENCODES

Calcineurin inhibitors (tacrolimus, cyclosporine) cause dose-dependent AKI primarily through afferent arteriolar vasoconstriction — a functional, often reversible drop in GFR resembling prerenal physiology, plus chronic interstitial fibrosis with prolonged use. They also cause hyperkalemia, hypomagnesemia, hypertension, and (cyclosporine) can trigger TMA. Check drug levels.

BOARD TRAP

In a transplant patient with rising creatinine, attributing it to rejection and increasing immunosuppression — high calcineurin-inhibitor levels cause vasoconstrictive AKI, so checking the trough level distinguishes drug toxicity (lower the dose) from rejection (intensify therapy).

7. Antivirals

WHAT YOU SEE

Tenofovir, cidofovir, foscarnet, 'pre-hydrate'

WHAT IT ENCODES

Antivirals injure the kidney by distinct mechanisms: tenofovir and cidofovir cause proximal tubulopathy/Fanconi syndrome (use probenecid + hydration with cidofovir); foscarnet causes ATN plus electrolyte derangements; acyclovir (IV, rapid infusion, volume-deplete) precipitates intratubular crystals causing obstructive AKI; indinavir/atazanavir cause crystal nephropathy and stones. Pre-hydration is the universal preventive.

BOARD TRAP

Giving rapid IV acyclovir without fluids — the trap is crystal-induced AKI from acyclovir, prevented by slow infusion and aggressive hydration; the discriminator is needle-shaped birefringent crystals on urine micro.

8. Probenecid / pentamidine

WHAT YOU SEE

PROBENECID and PENTAMIDINE bottles on antiviral shelf

WHAT IT ENCODES

Probenecid blocks tubular secretion of organic anions and is co-given with cidofovir to reduce its proximal-tubule uptake and nephrotoxicity. Pentamidine (for PCP) causes AKI plus hyperkalemia (it blocks the ENaC sodium channel like amiloride) and hypoglycemia/hypocalcemia. Both warrant close electrolyte and renal monitoring.

BOARD TRAP

Overlooking pentamidine's amiloride-like hyperkalemia — like trimethoprim, it blocks distal ENaC and raises potassium, a frequently tested mechanism that mimics a type 4 RTA picture.

9. Fanconi syndrome

WHAT YOU SEE

FANCONI SYNDROME central box, proximal tubular leaks

WHAT IT ENCODES

Fanconi syndrome is a generalized proximal tubule dysfunction causing urinary loss of glucose (glycosuria with normal serum glucose), phosphate (hypophosphatemia), bicarbonate (type 2 proximal RTA, non-anion-gap acidosis), amino acids (aminoaciduria), and uric acid. Drug causes include ifosfamide, tenofovir/cidofovir, expired tetracyclines, and heavy metals (lead, cadmium, mercury).

BOARD TRAP

Seeing glycosuria and assuming uncontrolled diabetes — in Fanconi the serum glucose is NORMAL; glycosuria with euglycemia plus phosphaturia and a non-gap acidosis points to proximal tubular dysfunction, not hyperglycemia.

10. Myoglobin

WHAT YOU SEE

MYOGLOBIN muscle figure, dipstick+ few RBCs

WHAT IT ENCODES

Rhabdomyolysis releases myoglobin, a heme pigment that causes ATN through tubular vasoconstriction, direct toxicity, and cast obstruction (worse in acidic urine). Labs: markedly elevated CK (>5000, often >15,000), hyperkalemia, hyperphosphatemia, hypocalcemia, and a urine dipstick positive for 'blood' with FEW RBCs on microscopy. Treat with aggressive IV fluids; consider urine alkalinization.

BOARD TRAP

A blood-positive dipstick with no/few RBCs on micro is misread as hematuria — it is pigment (myoglobin) cross-reacting with the dipstick, so confirm with CK and recognize pigment nephropathy rather than chasing a urologic bleed.

11. Hemoglobin

WHAT YOU SEE

HEMOGLOBIN red cell, dipstick+ few RBCs

WHAT IT ENCODES

Intravascular hemolysis (transfusion reactions, mechanical heart valves, PNH, severe malaria, G6PD crisis) releases free hemoglobin, a nephrotoxic pigment causing AKI by the same mechanisms as myoglobin. Clues: low haptoglobin, high LDH, indirect hyperbilirubinemia, schistocytes if mechanical/microangiopathic, and a blood-positive dipstick with few RBCs.

BOARD TRAP

Distinguishing hemoglobinuria from myoglobinuria on dipstick alone — both turn the dipstick positive for blood with few RBCs; the discriminator is plasma color (pink in hemoglobinemia, clear in rhabdo) plus CK (high only in rhabdo) and haptoglobin/LDH.

12. Uric acid / TLS

WHAT YOU SEE

URIC ACID crystal pot, 'up K up P up U, down Ca'

WHAT IT ENCODES

Tumor lysis syndrome (after chemo for bulky/high-turnover tumors like Burkitt, ALL) releases intracellular contents: uric acid and calcium-phosphate crystals precipitate in tubules causing obstructive AKI. The metabolic signature is hyperkalemia, hyperphosphatemia, hyperuricemia, and hypocalcemia (P binds Ca). Prevent with aggressive hydration plus rasburicase (high risk) or allopurinol (lower risk).

BOARD TRAP

Giving allopurinol alone to a high-burden patient already in TLS — allopurinol blocks new urate formation but does not clear existing urate; rasburicase (recombinant urate oxidase) actively degrades uric acid and is the answer for established/high-risk TLS (avoid in G6PD deficiency).

13. Myeloma cast

WHAT YOU SEE

Light chains, dipstick negative, MYELOMA CAST NEPHROPATHY

WHAT IT ENCODES

In multiple myeloma, filtered monoclonal free light chains (Bence Jones protein) bind Tamm-Horsfall protein and form obstructing casts (cast nephropathy / 'myeloma kidney'), the most common cause of myeloma AKI. The urine DIPSTICK is negative because it detects only albumin, not light chains — use the sulfosalicylic acid (SSA) test or urine protein electrophoresis. Hypercalcemia and contrast/NSAIDs precipitate it.

BOARD TRAP

Ruling out significant proteinuria because the dipstick is negative — the dipstick misses light chains entirely; the discriminator is a positive SSA test or UPEP showing an M-spike, and a dipstick-negative but SSA-positive urine is the classic myeloma tell.

14. NSAIDs / PPIs (AIN)

WHAT YOU SEE

PPI and NSAID culprits, AIN sick bay bed

WHAT IT ENCODES

NSAIDs and PPIs (plus beta-lactams, sulfonamides, rifampin, and allopurinol) cause acute interstitial nephritis, a T-cell-mediated hypersensitivity reaction days to weeks after exposure — distinct from direct ATN. NSAIDs additionally cause prerenal AKI (PG-mediated afferent constriction) and minimal-change disease/membranous nephropathy. Treatment of AIN is stopping the drug, with steroids if no recovery.

BOARD TRAP

Assuming AIN always follows a new drug within days — PPI- and NSAID-induced AIN can appear weeks to months after a chronically taken drug, so a 'stable home med' does not exclude it; the discriminator is sterile pyuria with WBC casts.

15. AIN: WBC casts/eos

WHAT YOU SEE

AIN SICK BAY: WBC casts + eosinophils, 'triad rarely present'

WHAT IT ENCODES

Acute interstitial nephritis is suggested by sterile pyuria, white-cell casts, and eosinophiluria, with peripheral eosinophilia and a mild rise in creatinine. The classic triad of fever, rash, and eosinophilia is present in only a minority (~10%) of cases, so its absence does NOT exclude AIN. Definitive diagnosis is biopsy; treat by withdrawing the offending drug.

BOARD TRAP

Excluding AIN because the patient lacks fever, rash, and eosinophilia — the full triad is rare; relying on it causes missed diagnoses, and WBC casts/sterile pyuria after a culprit drug are the more reliable clues.

16. Checkpoint inhibitors

WHAT YOU SEE

Checkpoint inhibitors box, AIN, ANA/ANCA

WHAT IT ENCODES

Immune checkpoint inhibitors (anti-PD-1/PD-L1 like nivolumab/pembrolizumab, anti-CTLA-4 like ipilimumab) cause immune-related AKI, most commonly acute interstitial nephritis, weeks to months into therapy and sometimes with positive ANA/ANCA. Management is holding the drug and high-dose corticosteroids, mirroring other immune-related adverse events.

BOARD TRAP

Attributing AKI in a cancer patient on immunotherapy to dehydration or contrast and continuing the drug — checkpoint-inhibitor AIN requires drug cessation plus steroids, and missing it allows progressive immune-mediated injury.